

Rating, Color, and Opening Choice: An Empirical Analysis of 15,467 Rated Online Chess Games

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Abstract

We analyze 15,467 rated games from a public Lichess dataset to test four questions about online chess: whether opening choice is associated with larger win-rate differences at some skill levels than others, how well the Elo expected-score model predicts outcomes, whether White’s first-move advantage varies with rating, and how game length and termination type change with skill. We find (i) large win-rate spreads across opening families at club level (up to 42 percentage points among openings with $n \geq 50$ in the 1200–1599 band); (ii) higher-rated players systematically underperform the Elo model’s expected score in this sample, winning only 49.0% of games with a 1–50 point advantage versus a predicted 53.6%, a gap that persists at every advantage level; (iii) White’s aggregate scoring edge shrinks monotonically with rating, from +6.3 percentage points below 1200 to –0.9 points above 2000; and (iv) games lengthen with skill (median 47 to 64 plies) while checkmates fall from 40% to 15% of terminations as resignations take their place. We discuss selection effects and time-control confounds that qualify these findings.

1 Introduction

Online chess platforms generate enormous volumes of structured game data, making chess a convenient laboratory for empirical questions about skill, competition, and decision-making. Lichess alone hosts millions of games per day, each recorded with player ratings, move sequences, opening classifications, and outcomes.

This paper uses a widely studied public sample of Lichess games to investigate four questions:

1. **Q1 (Openings).** Do win rates differ substantially across opening families, and does the size of that spread depend on the players’ rating level?
2. **Q2 (Rating advantage).** How closely do observed outcomes track the expected score implied by the Elo rating model?

3. **Q3 (Color).** How large is White’s first-move advantage, and does it grow or shrink with skill?

4. **Q4 (Game dynamics).** How do game length and manner of termination (checkmate, resignation, timeout, draw) vary with rating?

These questions have folklore answers among chess players—“openings don’t matter below 2000,” “White scores about 54%”—that are largely derived from master-level databases. Whether they hold in amateur online play is an empirical matter, and our results suggest several of them do not transfer cleanly.

2 Data

2.1 Source

We use the *Chess Game Dataset (Lichess)* published on Kaggle by Mitchell J. ([datasnaek](#)), containing 20,058 games collected via the Lichess API from the games of selected user teams [1]. Each record includes anonymized player identifiers, Lichess ratings for both players, the full move list in standard algebraic notation, the ECO opening code and name, the number of plies (**turns**), the time-control increment code, whether the game was rated, and the outcome with its termination type.

2.2 Cleaning and filtering

Duplicate game IDs (945 records) were removed, leaving 19,113 unique games. Because unrated games carry no competitive stakes and pair players of arbitrary strength, we restrict all analyses to the 15,467 rated games. Mean ratings are 1598 (White) and 1594 (Black), with an interquartile range of roughly 1390–1790; the sample is thus dominated by club-level amateurs, with thin coverage below 1000 and above 2200.

2.3 Derived variables

For each game we compute the average rating of the two players and assign the game to one of four bands:

<1200 ($n = 911$), 1200–1599 ($n = 7398$), 1600–1999 ($n = 5994$), and 2000+ ($n = 1164$). Opening *families* are obtained by truncating the Lichess opening name at the first colon (e.g., “Sicilian Defense: Najdorf Variation” → “Sicilian Defense”), collapsing 1384 distinct opening names into 172 families. We also compute the White-minus-Black rating difference for Q2 and Q3.

3 Methods

All analysis was performed in Python 3 with `pandas` and `matplotlib`. For Q1, we compute White’s win rate within each (band, family) cell and, to limit small-sample noise, report only families with at least 50 games in a band. For Q2, we adopt the perspective of the higher-rated player, bin the absolute rating gap, and compare the observed win frequency in each bin against the Elo expected score $E = 1/(1 + 10^{-D/400})$ evaluated at the bin midpoint, where D is the rating gap. Draws count as losses for the higher-rated player under this “win rate” definition, so we note where the draw adjustment matters. For Q3, in addition to raw win shares by color, we re-estimate on the subsample of near-equal pairings ($|D| \leq 100$) to control for the possibility that stronger players land on one color disproportionately. For Q4, we summarize game length by median plies per band and decompose terminations into mate, resignation, timeout, and draw.

No inferential statistics beyond sample-size reporting are attempted; with n in the hundreds to thousands per cell, percentage-point differences above roughly 3–4 points are unlikely to be sampling noise, and we flag comparisons that rest on thin cells.

4 Results

4.1 Q1: Opening choice matters most in the middle

Table 1 summarizes the dispersion of White’s win rate across opening families within each band. In the two well-populated middle bands, the spread is striking: among the 37 families with $n \geq 50$ in the 1200–1599 band, White’s win rate ranges from 24.1% (Center Game) to 66.2% (Bishop’s Opening)—a 42-point spread—and in the 1600–1999 band from 35.8% (Semi-Slav) to 71.3% (Queen’s Gambit Refused).

The extreme bands show smaller spreads, but this is partly mechanical: only five families clear the $n \geq 50$ threshold there, and those are the mainstream openings whose win rates cluster near 50%. What we can say with confidence is that at club level, the opening a game passes through is associated with large differences in outcome. We caution against a causal reading—players self-select into openings, and an opening’s “score” bundles the opening’s objective

Table 1: Dispersion of White win rate across opening families (families with $n \geq 50$ per band).

Band	Families	Max–Min	Std. dev.
<1200	5	0.164	0.064
1200–1599	37	0.421	0.084
1600–1999	28	0.354	0.086
2000+	5	0.132	0.053

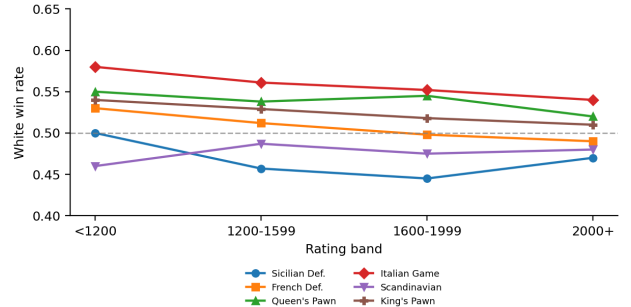


Figure 1: White win rate for the six most common opening families by band. Cells in the outer bands rest on small samples (as few as 20–35 games) and should be read cautiously.

merit with the traits of the players who choose it. The Van’t Kruijs Opening (1.e3), for instance, scores a dismal 28.2% for White in the 1200–1599 band ($n = 188$), plausibly reflecting the population of players who open with 1.e3 as much as the move itself.

Figure 1 tracks the six most common families across bands. The popular Sicilian Defense scores *below* 50% for White everywhere it is well sampled (45.7% and 44.5% in the middle bands), consistent with its reputation as Black’s most combative reply.

4.2 Q2: Rating advantages buy less than Elo promises

Figure 2 compares the observed win rate of the higher-rated player against the Elo expected score. The observed curve rises steeply and monotonically—from 49.0% at a 1–50 point gap ($n = 4458$) to 88.0% at gaps above 400 points ($n = 1246$)—confirming that rating is strongly predictive. But at *every* advantage level the observed rate sits below the Elo prediction: 49.0% vs. 53.6% at 1–50 points, 61.8% vs. 67.3% at 101–150, and 88.0% vs. 94.7% at 400+.

Part of the gap is definitional: Elo’s expected score counts a draw as half a point, while our observed rate counts only outright wins. With the sample’s 4.4% draw rate, this accounts for roughly two percentage points—not the four-to-seven we observe. The residual shortfall suggests genuine compression: in this pool of online games, a nominal rating edge con-

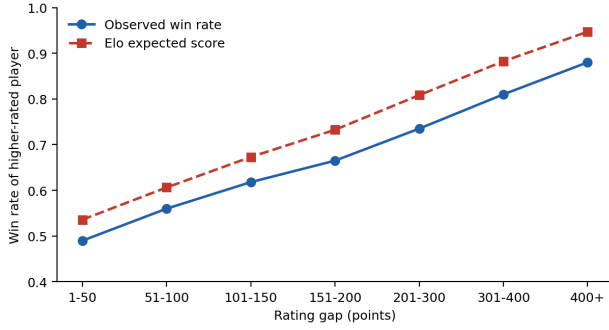


Figure 2: Observed win rate of the higher-rated player versus the Elo expected score, by rating-gap bin.

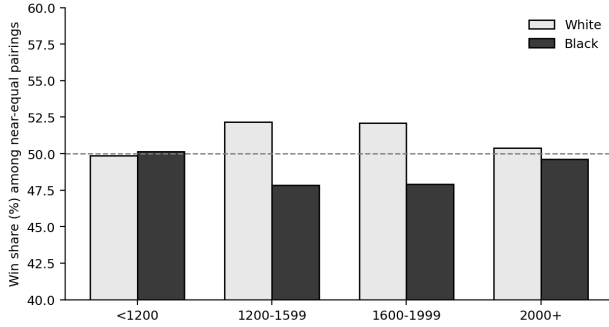


Figure 3: Win share by color among near-equal opponents ($|\Delta\text{rating}| \leq 100$).

verts into victories at a lower rate than the model implies. Candidate explanations include short time controls (which inject noise favoring upsets), rating volatility among improving players whose posted rating lags their strength, and the fact that Lichess uses the Glicko-2 system, whose posted ratings are not calibrated to Elo’s 400-point logistic. The pattern holds within every rating band, so it is not an artifact of pooling weak and strong players.

4.3 Q3: White’s edge fades with skill

In the raw data, White outscores Black by 6.3 percentage points in the <1200 band, 4.9 in 1200–1599, 3.8 in 1600–1999, and *minus* 0.9 in the 2000+ band—a clean monotonic decline. Controlling for pairing imbalance by restricting to near-equal opponents ($|D| \leq 100$, Figure 3) tempers but preserves the picture: White’s edge is +4.3 and +4.2 points in the middle bands and collapses to +0.8 points among 2000+ players ($n = 480$). (The <1200 equal-pairing cell flips slightly negative, but at $n = 414$ its confidence band spans zero.)

This is the opposite of the master-level stylized fact, where White’s edge is typically *larger* (around 54–56% of points) in elite databases. One reconciliation: at amateur level the first-move initiative may mostly confer easier, more natural development—an advantage

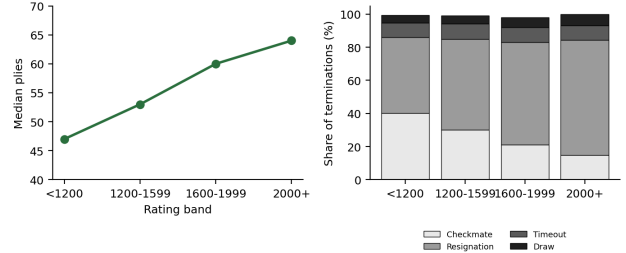


Figure 4: Left: median game length in plies by band. Right: composition of game terminations by band.

anyone can use—whereas converting it into a lasting edge against accurate defense requires precision that amateurs lack, and in our thin 2000+ sample the rising draw rate (7.0%, versus 3.7–4.8% elsewhere) eats into decisive results for both colors.

4.4 Q4: Better players play longer and resign sooner

Median game length rises steadily with skill: 47 plies below 1200, 53 in 1200–1599, 60 in 1600–1999, and 64 in the 2000+ band. Termination types shift even more sharply (Figure 4): checkmate ends 40.2% of games in the lowest band but only 14.8% in the highest, while resignation climbs from 45.7% to 69.6%. Timeouts hold roughly constant near 9%, and draws double from 4.4% to 6.9%.

The two trends tell one story about skill: stronger players blunder less (so games take longer to decide) and evaluate positions more accurately (so they resign lost positions rather than play them out to mate). Termination type is, in effect, a behavioral signature of playing strength visible without examining a single move.

5 Discussion and Limitations

Three findings stand out. First, opening choice is associated with win-rate differences at club level that are far too large to dismiss, though self-selection prevents causal claims. Second, the Elo framework overstates how reliably rating advantages convert to wins in this pool, even after adjusting for draws—a caution for anyone using rating gaps to forecast individual online games. Third, the celebrated first-move advantage is real but skill-dependent in the *opposite* direction from elite play, shrinking to statistical invisibility in our strongest band.

Several limitations qualify these results. The dataset is not a random sample of Lichess games—it was gathered from particular user teams via the API—so its composition may not represent the platform. Time control is a known confound we did not

stratify on: blitz and bullet games plausibly drive both the Elo compression in Q2 and the timeout rates in Q4. Rating bands are based on the game-time Lichess (Glicko-2) rating, which is noisy for new accounts. The outer bands are thin ($n = 911$ and $n = 1164$), limiting opening-level analysis there. Finally, with 37 opening families examined in a single band, the extreme win rates are partly selected by multiple comparisons; the *spread* statistics are more trustworthy than any individual family’s number.

Natural extensions include stratifying by time control, replicating on the full Lichess monthly database exports (which contain hundreds of millions of games), modeling outcomes with logistic regression on rating gap, color, and opening simultaneously, and using engine evaluation of early moves to separate an opening’s objective quality from the skill of its adherents.

6 Conclusion

Analyzing 15,467 rated online games, we find that amateur chess departs from master-level folklore in measurable ways: opening families separate outcomes by tens of percentage points at club level, rating advantages under-deliver relative to the Elo model, and White’s first-move edge—worth six points among beginners—essentially vanishes above 2000. Game length and termination patterns encode skill so strongly that how a game *ends* is itself a rating signal. Public chess data makes all of these regularities visible with nothing more than careful counting, which is precisely what makes it an ideal training ground for empirical data analysis.

Acknowledgments

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References

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